

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2024

Programme: BE

Full Marks:100

Course: Engineering Economics (New)

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) List out the principles of Economy. 5
b) How you understand supply and demand. 5
c) Describe about manufacturing cost and Non-manufacturing cost. 5
2. a) A man is planning to retire in 25 years. He wishes to deposit regular money every 3 months until he retires so that he will receive annual payments of Rs.4, 50,000 after the first year of his retirement for the next 10 years. How much must he deposit if the interest rate is 8%, compounded monthly? 8
b) Explain about simple interest, compound interest, nominal interest and effective interest in your own word. 7
3. a) Calculate the simple and discounted payback period for the project when MARR is 20% 8
Initial investment: Rs 25000
Annual Revenue: Rs 8000
Salvage Value: Rs 5000
Expected Life: 5 year.
b) Compute IRR and suggest the investment decision for the project given below: 7
Initial investment: Rs 90000
Annual Revenue: Rs 27000
Annual Expenses: Rs 8000
Salvage Value: Rs 10000
Expected Life: 6 year
MARR: 15%

4. a) Using co-terminated assumption recommend the best project taking study period as 5 years 8

Project	A	B
Initial Investment (Rs)	3,50,000	5,00,000
Annual Revenues (Rs)	1,30,000	1,75,000
Annual cost (Rs)	15,000	25,000
Salvage value (Rs)	35,000	50,000
Useful life	5 years	8 years
MARR	10%	

OR

Evaluate the following two feasible investment A and B having different useful lives, if MARR is 10% per year. Use PW method with repeatability assumption.

<u>Project: A</u> Initial investment: Rs 50000 Annual Revenue: Rs 25000 Annual Expenses: Rs 3000 Salvage Value: Rs 15000 Expected Life: 3 year	<u>Project: B</u> Initial investment: Rs 150000 Annual Revenue: Rs 70000 Annual Expenses: Rs 2000 Salvage Value: Rs 40000 Expected Life: 5 year
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- b) Suppose a company has decided to purchase a new machine having initial cost of Rs 18,000 by expecting 8 years useful life & have operating cost of Rs 3000 in 1st year. For remaining years, operating cost increase each year by 15% over previous year's operating cost & salvage value declines each year by 20% from the previous year's salvage value. Find the economic service life of this new machine when MARR is 12% per year. 7
5. a) Perform Sensitivity analysis using PW method over a range of (+ or -) 20% in initial investment, net annual revenue and salvage value. 8
- Initial investment: Rs 200000
Annual Revenue: Rs 50000
Annual Expenses: Rs 5000
Salvage Value: Rs 25000
Expected Life: 10 year
MARR: 12% per year

- b) What do you understand about break-even analysis? State the procedures to identify break even point. 7

OR

If you purchase Jagadamba cement at the rate Rs 650 and selling at the rate of Rs 900, you are paying shop rent change Rs 10000 and Rs 12000 for the helper salary. What is the break-even quantity that you must sold. Let us say you have 10000 bags of cement is in stock.

6. a) A grinding machine has initial cost & annual saving per year of Rs. 40,000 & Rs. 20,000 respectively with annual O & M cost of Rs. 6,000. It will depreciate by MACRS method & will have salvage value of Rs. 4,000 at the end of 5th year. Estimate BTCF & ATCF. The company pays income tax @ 40%. Evaluate the profitability by FW method if interest rate is 12%. Also, calculate the interest rate by IRR analysis. 8
- b) Discuss about different methods of financing including its merits and demerits. 7
7. Write short notes on: (**Any two**) 2×5
- a) Inflation
 - b) Risk Analysis
 - c) Independent and contingent project